PONTIFICIA UNIVERSIDAD CATÓLICA DE VALPARAÍSO

Computational Optimization — Ph.D. Coding Exam

Production planning with inventory — implementation and modelling in Python + Gurobi

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Name: ______________________ **Date:** __________ **Total:** 100 pts (Part A: 80 + Part B: 20)

Deliverables

- Two Python scripts: `part_A_<lastname>.py` and `part_B_<lastname>.py`, runnable with `python3` (no external data files).
- All figures generated by your code as separate `.png` files.
- A short PDF report (3–5 pages) summarizing your results and answering the discussion questions.
- In Part B, the mathematical formulation of the new constraint must appear in your report (LaTeX or scanned hand-writing is acceptable).

Allowed packages

`gurobipy`, `numpy`, `matplotlib`, `pandas`. No other optimization libraries.

PART A — Implementation of the Given Model (80 pts)

A specialty steel manufacturer plans production over $|T| = 4$ monthly periods using $|M| = 2$ parallel furnaces (F1, F2), producing $|K| = 3$ products (A, B, C) and shipping to $|J| = 2$ distribution centers (DC1, DC2). The mathematical model is given below. You must implement it in Gurobi exactly as written.

Context and key concepts

Before diving into the math, here is what each concept means *operationally*. All of these will appear as variables, parameters, or constraints in the model:

- **Production.** Each furnace m can manufacture units of any product k in any period t. The number of units produced is denoted $x_{m,k,t}$, and incurs a unit production cost $CP_{m,k,t}$. Producing one unit of k consumes $a_{m,k}$ machine-hours on furnace m.
- **Furnace activation.** A furnace is either ON ($z_{m,t} = 1$) or OFF ($z_{m,t} = 0$) in a given period. Turning a furnace on costs a fixed activation fee $CF_{m,t}$ (electricity, ramp-up, staffing) that is paid even if production is small. When OFF, no production and no overtime are possible.
- **Regular and overtime hours.** When ON, a furnace has up to $H_{m,t}$ regular hours and an additional $H^{ov}_{m,t}$ overtime hours. Overtime hours $o_{m,t}$ are billed at the premium rate $c^{ov}_{m,t}$ per hour (think: extra labor cost).
- **Deliveries to DCs.** Production is shipped to distribution centers as deliveries $y_{j,k,t}$. Each shipment carries a unit transportation cost $CT_{j,k,t}$ that depends on the destination DC. Deliveries are how demand is actually *satisfied*.

- **Inventory (holding).** If the plant produces more than is delivered in a period, the surplus is stored as physical inventory $I_{k,t}$ at the end of period t, carrying a unit **holding cost** CH_k per period (think: warehouse rental, capital tied up in stock). A finite cap S_k limits how much can be stored.
- **Back-orders (unmet demand).** Conversely, if the plant cannot meet all current-period demand, the shortfall becomes a **back-order** $B_{k,t}$ – a promise to deliver later. Each unit of back-order carries a penalty CB_k per period it remains unfilled (think: customer dissatisfaction, late-delivery fee). Back-orders accumulate across periods until cleared by a future delivery. At the end of the horizon, all back-orders must be zero (no unfilled promises left).
- **Service level.** A regulatory or commercial constraint requires that at least a fraction ρ_k of current-period demand be delivered *within the same period* (i.e., not back-ordered). This prevents the model from systematically back-ordering everything cheap to back-order.
- **Production smoothing.** Furnaces and supply chains do not like abrupt changes: the total production of any product cannot jump by more than $\pm\epsilon = 25\%$ from one period to the next. This couples the time periods and prevents extreme "bang-bang" policies (full production one period, idle the next).

The objective minimizes the sum of all six cost components: production + transport + activation + overtime + holding + back-order.

Sets

$K = \{A, B, C\}$, $M = \{F1, F2\}$, $J = \{DC1, DC2\}$, $T = \{1, 2, 3, 4\}$.

Parameters (instance)

- Demand $D_{j,k,t}$ is given as a fixed table (see Table 1 below).
- Production cost $CP_{m,k,t} = c_0^m + c_0^k$ with $c_0^{F1} = 8,\ c_0^{F2} = 10$ and $c_0^A = 0,\ c_0^B = 1,\ c_0^C = 2$.
- Transportation cost $CT_{j,k,t} = 1$ for DC1, 2 for DC2.
- Holding cost $CH_k = (1.0,\ 1.2,\ 0.8)$ for (A, B, C). Back-order cost $CB_k = (4,\ 5,\ 3)$.
- Machine-hours per unit: $a_{m,k} = 0.5,\ 0.7,\ 0.4$ for $k = A, B, C$ (same for both furnaces).
- Regular hours $H_{m,t} = 50$, max overtime $H^{ov}_{m,t} = 20$, activation cost $CF_{m,t} = 200$, overtime cost $c^{ov}_{m,t} = 15$.
- Initial inventory $I_k^0 = 0$, inventory cap $S_k = 200$, service-level $\rho_k = 0.60$.
- Smoothing tolerance $\epsilon = 0.25$.

Table 1: **Demand** $D_{j,k,t}$ in units (you must hard-code these values into a Python dictionary).

		DC1				DC2			
		$t=1$	$t=2$	$t=3$	$t=4$	$t=1$	$t=2$	$t=3$	$t=4$
Product	A	28	27	33	30	27	30	28	33
	B	17	20	26	22	19	19	26	24
	C	11	17	14	17	13	19	17	18

Decision variables

- $x_{m,k,t} \geq 0$: units of k produced on furnace m in period t.
- $y_{j,k,t} \geq 0$: units of k delivered to DC j in period t.
- $I_{k,t} \geq 0$: physical inventory of k at end of period t.
- $B_{k,t} \geq 0$: cumulative back-orders of k at end of period t.
- $o_{m,t} \geq 0$: overtime hours of furnace m in period t.
- $z_{m,t} \in \{0,1\}$: 1 if furnace m is ON in period t.

Mathematical formulation

$$
\begin{aligned}
\min \quad & \sum_{m,k,t} CP_{m,k,t}\, x_{m,k,t} \;+\; \sum_{j,k,t} CT_{j,k,t}\, y_{j,k,t} \;+\; \sum_{m,t} CF_{m,t}\, z_{m,t} \\
& + \sum_{m,t} c^{ov}_{m,t}\, o_{m,t} \;+\; \sum_{k,t} CH_k\, I_{k,t} \;+\; \sum_{k,t} CB_k\, B_{k,t} && (1)
\end{aligned}
$$

$$
\begin{aligned}
\text{s.t.} \quad & I_{k,t} \;=\; I_{k,t-1} + \sum_m x_{m,k,t} - \sum_j y_{j,k,t}, && \forall k,t \quad (I_{k,0} = I^0_k) && (2)\\
& \sum_j y_{j,k,t} + B_{k,t} \;=\; \sum_j D_{j,k,t} + B_{k,t-1}, && \forall k,t \quad (B_{k,0} = 0) && (3)\\
& B_{k,|T|} \;=\; 0, && \forall k && (4)\\
& \sum_j y_{j,k,t} \;\geq\; \rho_k \sum_j D_{j,k,t}, && \forall k,t && (5)\\
& I_{k,t} \;\leq\; S_k, && \forall k,t && (6)\\
& \sum_k a_{m,k}\, x_{m,k,t} \;\leq\; H_{m,t}\, z_{m,t} + o_{m,t}, && \forall m,t && (7)\\
& o_{m,t} \;\leq\; H^{ov}_{m,t}\, z_{m,t}, && \forall m,t && (8)\\
& X_{k,t} \;\leq\; (1+\epsilon)\, X_{k,t-1}, && \forall k,\; t \geq 2 && (9)\\
& X_{k,t} \;\geq\; (1-\epsilon)\, X_{k,t-1}, && \forall k,\; t \geq 2 && (10)
\end{aligned}
$$

where $X_{k,t} := \sum_m x_{m,k,t}$.

Part A tasks

(A.1) (25 pts) Implement and solve. Build the MILP in Gurobi exactly as specified. Solve and report: (i) the optimal objective value; (ii) a table showing $D_{k,t}, X_{k,t}, Y_{k,t}, I_{k,t}, B_{k,t}$ for every (k,t); (iii) a table showing for every (m,t) whether furnace m is ON ($z_{m,t}$) and how many regular/overtime hours are used.

(A.2) (20 pts) Holding-cost sensitivity. Introduce a multiplier $\alpha \geq 0$ and replace $CH_k \to \alpha\, CH_k$. Sweep $\alpha \in [0,5]$ with at least 20 grid points and re-solve at each value. Produce a plot showing total cost and total end-of-period inventory $\sum_{k,t} I^*_{k,t}$ vs. α on a twin-axis figure. Identify the value of α at which the optimizer abandons the "produce-ahead-and-store" strategy.

(A.3) (20 pts) Overtime-cost sensitivity. Introduce a multiplier β and replace $c^{ov}_{m,t} \to \beta\, c^{ov}_{m,t}$. Sweep $\beta \in [0.5, 4.0]$ with at least 15 grid points. Plot total cost and total overtime hours $\sum_{m,t} o^*_{m,t}$ vs. β. Identify the breakeven β at which the optimizer stops using overtime altogether, and interpret economically.

(A.4) (15 pts) Discussion. In your report, answer in a few sentences each:

1. Why does the holding-cost curve show a clear "kink" rather than a smooth slope?
2. In the base solution, is furnace F2 used in all periods or only in some? Why?
3. What is the role of the smoothing constraints (9)–(10) and how would the solution change qualitatively if you removed them?

PART B — New Constraint: Minimum Lot Size + Setup Cost (20 pts)

The plant manager informs you of a new operational rule that must be added to the model:

> *"Due to thermal start-up considerations, whenever a furnace m produces* ***any*** *units of product k in period t, the furnace must produce* ***at least*** *L_k units of that product. Additionally, each such product-furnace-period combination triggers a fixed setup cost $SC_{m,k}$ (covering retooling, calibration and quality control). If the furnace does not produce product k in period t, neither the minimum lot nor the setup cost applies."*

New parameters

- Minimum lot per product: $L_k = (25, 20, 15)$ for $k = A, B, C$.
- Setup cost: $SC_{m,k} = 50$ for all (m, k).
- **Big-M value (given to you).** Whenever you need a Big-M in your formulation, use

$$M_{m,k,t} \; = \; \frac{H_{m,t} + H^{ov}_{m,t}}{a_{m,k}}.$$

  With the instance parameters $H_{m,t} = 50$, $H^{ov}_{m,t} = 20$, this gives $M_{m,A,t} = 140$, $M_{m,B,t} = 100$, $M_{m,C,t} = 175$. You are NOT asked to justify this value; just use it.

Part B tasks

(B.1) (10 pts) Mathematical formulation. Translate the new rule into linear constraints. Specifically:

(i) Define a new family of binary decision variables (with explicit domain and verbal definition) that captures whether "furnace m produces product k in period t".

(ii) Write the two linear inequalities that couple your new variables to the existing $x_{m,k,t}$. Label each side: the *Big-M side* (forcing $x = 0$ when the binary is 0) and the *minimum-lot side* (forcing $x \geq L_k$ when the binary is 1). Use the $M_{m,k,t}$ given above.

(iii) Modify the objective function to include the new setup cost.

This part must appear in your **written report** (LaTeX or hand-written). *Do not jump straight to code without writing the formulation first.*

(B.2) (7 pts) Implementation and comparison. Add your new variables and constraints to the Part A Gurobi model and re-solve. Report: (i) the new optimal objective value; (ii) the number of (m, k, t) setups paid; (iii) a side-by-side comparison table of $X^{(A)}_{m,k,t}$ vs. $X^{(B)}_{m,k,t}$ for every (m, k, t), highlighting where the production plan changes between Part A and Part B.

(B.3) (3 pts) Discussion. Briefly answer (2–3 sentences each):

1. Why does the cost in Part B exceed the cost in Part A? Identify the two distinct sources of the cost increase.
2. Suppose the setup cost $SC_{m,k}$ becomes extremely large. Would the number of setups paid necessarily decrease in this instance? Use the structure of the existing model (especially the smoothing constraints) to explain.

— END OF EXAM —